

Ryan Cho

Senior Full-Stack Software Engineer

+1 608 360 1459 · Ryanmcho100@gmail.com · <https://www.linkedin.com/in/ryan-cho-2020/> · <https://github.com/Chomama> · New York, United States

PROFESSIONAL SUMMARY

Full-stack engineer with 7+ years building scalable web applications and cloud-native services in regulated financial and enterprise environments. Proficient in React, TypeScript, Java, and Python across RESTful microservices, event-driven pipelines, and secure, access-controlled systems. Delivers search, data processing, and AI-enabled capabilities through collaboration with product, UX, and domain stakeholders. Mentors engineers, leads architecture reviews, and ships accessible, high-performance interfaces.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and delivered **full-stack platform features** for an internal investment analytics system serving **40+ portfolio managers**, combining **React and TypeScript** frontends with **Java microservices** and RESTful APIs.
- Designed **event-driven data ingestion pipelines** using **Apache Kafka** and RabbitMQ to process market and portfolio events, reducing downstream reporting latency by **35%** across distributed services.
- Built **Python backend services** for document indexing and **semantic search**, enabling analysts to retrieve structured insights from large financial datasets with **sub-second query response** times.
- Implemented **role-based access control**, encryption at rest and in transit, and audit logging aligned with **regulatory compliance** requirements for sensitive financial data handling.
- Containerized services with **Docker** and deployed to **AWS and Azure** using **Kubernetes** orchestration, establishing CI/CD pipelines that shortened release cycles from weekly to **daily deployments**.
- Partnered with product managers and UX designers to deliver **responsive, accessible** React applications, applying performance tuning that cut initial page load times by **28%** across primary workflows.
- Integrated **generative AI-assisted summarization** into research workflows, connecting LLM outputs to existing APIs and validation layers for controlled production rollout.
- **Mentored three junior engineers** through code reviews, pairing sessions, and architecture discussions, elevating team test coverage and establishing shared patterns for microservice development.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **RESTful APIs and Java backend services** supporting trade reporting and operational dashboards used by **200+ internal analysts** across global markets divisions.
- Built **React and JavaScript** front-end modules for compliance tooling, delivering **responsive interfaces** that streamlined manual review workflows and reduced processing time by **20%**.
- Implemented **event-driven messaging** with Kafka for transaction data propagation between legacy and modernized systems, ensuring reliable delivery across **15+ downstream consumers**.
- Applied **secure coding practices** including input validation, encryption, and access controls to meet banking security and **privacy standards** for customer-facing applications.
- Optimized **SQL data models** and database queries for reporting pipelines, improving batch job performance and enabling faster retrieval across **multi-terabyte datasets**.
- Participated in **architecture reviews** and technical design discussions, contributing patterns for microservice decomposition and API versioning across cross-functional teams.
- Collaborated with QA and business stakeholders using **Cypress** and Jest test suites, increasing regression coverage for critical payment and reporting workflows.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Contributed to **React and JavaScript** features for a tablet-based restaurant check management application deployed across **50+ client locations**.
- Assisted in building **Node.js REST APIs** connecting mobile clients to PostgreSQL data stores, supporting real-time status updates for front-of-house operations.
- Wrote **Cypress end-to-end tests** covering checkout and inventory flows, helping the team catch regressions before each sprint release.
- Refactored legacy UI components to improve **responsive design** and cross-browser compatibility on iPad and mobile form factors.
- Participated in agile ceremonies and **code reviews**, gaining hands-on experience with Git workflows and collaborative software development practices.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **HTML5, CSS, and JavaScript** pages for a nonprofit web presence, delivering accessible layouts aligned with organizational branding guidelines.
- Implemented **responsive design** patterns ensuring consistent experiences across desktop and mobile devices for community-facing content.
- Integrated third-party forms and analytics into the site while maintaining **performance optimization** for low-bandwidth users.
- Collaborated with stakeholders to translate content requirements into maintainable front-end templates deployed via shared hosting infrastructure.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Redux, React Query, Responsive Design, Accessibility, Jest, Cypress

Backend: Java, Python, Node.js, RESTful APIs, Microservices, Object-Oriented Programming, Design Patterns, Event-Driven Architecture

Databases: PostgreSQL, MySQL, Oracle, MongoDB, Redis, Data Modeling, SQL Optimization

Cloud & DevOps: AWS, Microsoft Azure, Google Cloud Platform, Docker, Kubernetes, CI/CD, Infrastructure as Code

Monitoring & Tools: Apache Kafka, RabbitMQ, Apache Flink, Git, Grafana, Logging & Observability, API Testing

Workflow & PM: Agile, Scrum, Jira, Code Review, Technical Leadership, Mentorship, Cross-Functional Collaboration